Historical per-map projection derivation. The three-projection ABBA4 implementation described in this note has been removed. Current `ABBA4Implicit` uses one outer projection around the complete unprojected composition. See the canonical theory and the six-phase simulation diagram for the implemented method.

ABBA4 Implicit: Simultaneous State–Multiplier Formulation

Three signed simultaneous solves in the fourth-order triple jump

Physical GC2D branch with one diagonal constraint per projected substep

1 Purpose and scope

The `simultaneous_state_multiplier` option of `ABBA4Implicit` computes the same ideal physical map as the `reduced_multiplier` option, but it retains the duplicated output state as part of each nonlinear solve. One fourth-order step contains three independent projected ABBA substeps. Consequently, for one guiding-centre particle, it solves three 6×6 nonlinear systems rather than three 2×2 reduced Schur-complement systems.

This note describes the physical-state branch. Setting An `EnergyObserver` uses the same physical projected solves and adds a triangular auxiliary update of $\kappa = k/2$ from the accepted stages; it does not change the dimensions below. The `fully_extended` branch duplicates the complete autonomous state and therefore uses different dimensions.

The two physical formulations agree whenever each projection root is locally unique and both nonlinear algorithms converge to that root. Finite stopping tolerances and floating-point linear solves can produce round-off-level differences.

2 Fourth-order signed composition

Define the triple-jump coefficients

$$\gamma = \frac{1}{2 - 2^{1/3}}, \qquad \delta = -\frac{2^{1/3}}{2 - 2^{1/3}}, \qquad (c_1, c_2, c_3) = (\gamma, \delta, \gamma).$$

They satisfy

$$c_1 + c_2 + c_3 = 1, \qquad c_1^3 + c_2^3 + c_3^3 = 0.$$

For an outer step $(t_n, z_n) \mapsto (t_n + h, z_{n+1})$, set

$$\tau_1 = t_n, \qquad \tau_2 = t_n + c_1 h, \qquad \tau_3 = t_n + (c_1 + c_2)h,$$

and

$$\ell_j = c_j h, \qquad \tau_j^+ = \tau_j + \ell_j.$$

The chronological physical updates are

$$z^{[0]} = z_n, \qquad z^{[j]} = \Psi^{[2]}_{\ell_j, \tau_j}(z^{[j-1]}), \quad j = 1, 2, 3, \qquad z_{n+1} = z^{[3]}.$$

The middle duration $\ell_2 = \delta h$ is negative. Its endpoint precedes its start, but the same endpoint-time formulas remain valid without a special backward branch.

3 Duplicated state and one signed ABBA map

The physical state $z = (X, Y) \in \mathbb{R}^2$ satisfies

$$\dot{z} = f(z,t), \qquad f(z,t) = J_0 \nabla_z \phi(z,t), \qquad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Its spatial Jacobian is

$$W(z,t) = D_z f(z,t) = J_0 \nabla_z^2 \phi(z,t).$$

Introduce the duplicated state and the diagonal and correction maps

$$\mathcal{Y} = (u, v) \in \mathbb{R}^4, \qquad G = (I_2 \ -I_2), \qquad N = G^T = \begin{pmatrix} I_2 \\ -I_2 \end{pmatrix}, \qquad E = \begin{pmatrix} I_2 \\ I_2 \end{pmatrix}.$$

Thus $Ez = (z, z)$ lies on $\mathcal{M} = \{(u, v) : u = v\}$, $N\mu = (\mu, -\mu)$ separates the copies, and

$$GE = 0, \qquad GN = 2I_2.$$

For substep j, let $s_j = \ell_j/2$. Starting from $(u_j^{(0)}, v_j^{(0)})$, the signed endpoint-time ABBA map is

$$\begin{aligned} u_j^{(1)} &= u_j^{(0)} + s_j f(v_j^{(0)}, \tau_j), \\ v_j^{(1)} &= v_j^{(0)} + s_j f(u_j^{(1)}, \tau_j), \\ v_j^{(2)} &= v_j^{(1)} + s_j f(u_j^{(1)}, \tau_j^+), \\ u_j^{(2)} &= u_j^{(1)} + s_j f(v_j^{(2)}, \tau_j^+). \end{aligned}$$

We denote the resulting duplicated map by

$$\Phi^{\mathrm{ABBA}}_{\ell_j, \tau_j}(u_j^{(0)}, v_j^{(0)}) = (u_j^{(2)}, v_j^{(2)}).$$

At its four stage points define

$$W_{j,1} = W(v_j^{(0)}, \tau_j), \quad W_{j,2} = W(u_j^{(1)}, \tau_j), \quad W_{j,3} = W(u_j^{(1)}, \tau_j^+), \quad W_{j,4} = W(v_j^{(2)}, \tau_j^+),$$

and $S_j = W_{j,2} + W_{j,3}$. The exact duplicated-space Jacobian is

$$J_{\mathrm{ABBA},j} = \begin{pmatrix} I_2 + s_j^2 W_{j,4} S_j & s_j(W_{j,1} + W_{j,4}) + s_j^3 W_{j,4} S_j W_{j,1} \\ s_j S_j & I_2 + s_j^2 S_j W_{j,1} \end{pmatrix}.$$

The factor order is essential because the stage Jacobians do not commute in general.

4 Simultaneous equations for substep j

Given $z^{[j-1]}$, pre-correct the diagonal input with an independent multiplier μ_j:

$$\widehat{\mathcal{Y}}_j(\mu_j) = Ez^{[j-1]} + N\mu_j = (z^{[j-1]} + \mu_j, z^{[j-1]} - \mu_j).$$

The simultaneous unknown is

$$(\mathcal{Y}_j^+, \mu_j) \in \mathbb{R}^4 \times \mathbb{R}^2, \qquad \mathcal{Y}_j^+ = (y_{u,j}, y_{v,j}).$$

It is determined by

$$d_j(\mathcal{Y}_j^+, \mu_j) := \mathcal{Y}_j^+ - N\mu_j - \Phi^{\mathrm{ABBA}}_{\ell_j, \tau_j}(Ez^{[j-1]} + N\mu_j) = 0,$$

$$g_j(\mathcal{Y}_j^+) := G\mathcal{Y}_j^+ = 0.$$

In component blocks, the residual is

$$d_{u,j} = y_{u,j} - \mu_j - u_j^{(2)}(\mu_j), \qquad d_{v,j} = y_{v,j} + \mu_j - v_j^{(2)}(\mu_j), \qquad g_j = y_{u,j} - y_{v,j}.$$

At convergence the accepted physical state is represented neutrally by

$$z^{[j]} = \frac{y_{u,j} + y_{v,j}}{2}.$$

The mean removes the antisymmetric component left by finite stopping tolerance; at the exact root both output copies are identical.

5 Exact simultaneous Newton system

Because G and N are constant,

$$D_{\mathcal{Y}_j^+} d_j = I_4, \qquad D_{\mu_j} d_j = -(I_4 + J_{\mathrm{ABBA},j})N,$$

$$D_{\mathcal{Y}_j^+} g_j = G, \qquad D_{\mu_j} g_j = 0.$$

Each Newton correction therefore solves

$$\boxed{\begin{pmatrix} I_4 & -(I_4 + J_{\mathrm{ABBA},j})N \\ G & 0 \end{pmatrix} \begin{pmatrix} \Delta\mathcal{Y}_j^+ \\ \Delta\mu_j \end{pmatrix} = -\begin{pmatrix} d_j \\ g_j \end{pmatrix}.} \tag{1}$$

The update adds $(\Delta\mathcal{Y}_j^+, \Delta\mu_j)$ to the current iterate. No matrix inverse is formed: the implementation solves one dense 6×6 system per independent particle.

The initial iterate is

$$\mu_j^{(0)} = 0, \qquad \mathcal{Y}_j^{+,(0)} = \Phi_{\ell_j,\tau_j}^{\mathrm{ABBA}}(Ez^{[j-1]}).$$

It makes $d_j = 0$ initially, while the generally nonzero copy difference g_j drives the first correction.

For

$$\sigma_j = \max\left(1, \left\|z^{[j-1]}\right\|_\infty\right),$$

the solve stops when

$$\max\left(\|d_j\|_\infty, \|g_j\|_\infty\right) \le \mathrm{atol} + \mathrm{rtol}\,\sigma_j.$$

Each of the three substeps has its own scale, iteration count, and stopping decision.

6 Schur complement and reduced-formulation equivalence

Eliminating $\Delta\mathcal{Y}_j^+$ from (1) produces

$$K_j = G(I_4 + J_{\mathrm{ABBA},j})G^T = 2I_2 + GJ_{\mathrm{ABBA},j}G^T.$$

The reduced residual is

$$R_j(\mu_j) = G\Phi_{\ell_j,\tau_j}^{\mathrm{ABBA}}(Ez^{[j-1]} + G^T\mu_j) + 2\mu_j,$$

and $K_j = D_{\mu_j} R_j$. Moreover,

$$Gd_j = g_j - R_j(\mu_j).$$

Substitution into the second block row of (1) gives

$$K_j\Delta\mu_j = -R_j(\mu_j),$$

which is exactly the reduced Newton correction. With equal initial multipliers, the two formulations therefore generate the same multiplier iterates in exact arithmetic. The simultaneous branch additionally updates the duplicated output workspace.

This equivalence is local to each signed substep. ABBA4 still requires three different roots because $(z^{[j-1]}, \tau_j, \ell_j)$ changes with j.

7 Broyden and multiple-particle layout

Selecting `nonlinear_solver="broyden"` preserves the simultaneous residual (d_j, g_j) but replaces the exact Newton matrix by a good rank-one secant update. The initial approximation has the same block structure as (1) with the unprojected Jacobian approximated by the identity. Broyden changes the root-finding algorithm, not the defining projected map.

For N_p independent guiding-centre particles, accepted physical states use component-major storage,

$$(X_1, \dots, X_{N_p}, Y_1, \dots, Y_{N_p}).$$

Newton reshapes this storage into N_p independent 6×6 blocks and packs the corrections back into component-major order. Broyden instead treats the complete simultaneous vector as one system of dimension $6N_p$.

8 Symmetry and symplecticity of the complete map

At an exact root, every substep satisfies

$$\mathcal{Y}_j^+ - N\mu_j = \Phi^{\mathrm{ABBA}}_{\ell_j,\tau_j}(Ez^{[j-1]} + N\mu_j), \qquad \mathcal{Y}_j^+ \in \mathcal{M}.$$

The endpoint-time ABBA map is self-adjoint under a signed-step reversal. The symmetric projection then gives

$$\Psi^{[2]}_{-\ell_j,\tau_j^+} \circ \Psi^{[2]}_{\ell_j,\tau_j} = I.$$

Because (γ, δ, γ) is palindromic, the complete composition obeys

$$\Psi^{[4]}_{-h,t_n+h} \circ \Psi^{[4]}_{h,t_n} = I.$$

Each elementary ABBA shear preserves the duplicated form

$$\Omega_\times = \begin{pmatrix} 0 & J_0 \\ J_0 & 0 \end{pmatrix}.$$

Hairer's symmetric projection restricts that preservation to the physical diagonal, so the ideal physical Jacobian J_j of each signed substep satisfies

$$J_j^T J_0 J_j = J_0.$$

The outer Jacobian is $J_3 J_2 J_1$ and is therefore symplectic as well. In finite arithmetic, both reversibility and symplecticity include the defects of all three nonlinear solves.

9 Implementation correspondence

The production path mirrors the derivation:

1. `order4_implicit.py` creates the three signed durations and passes the accepted physical output of one solve into the next;
2. `_projection_simultaneous.py` evaluates (d_j, g_j) and assembles the particle-wise matrix (1);
3. `_projection_common.py` evaluates the endpoint-time stages and $J_{\mathrm{ABBA},j}$;
4. each accepted outer step stores three ordered substep observations and three columns of nonlinear diagnostics;
5. aggregate diagnostics sum iteration and residual-evaluation counts, retain the worst normalized residual, and record the maximum multiplier norm.

The three simultaneous systems are solver workspaces. They are not trajectory states and do not introduce a fourth composition stage or a new public method.